Study design and information exposure

One source-reviewed freeze; complete systems, annotation budgets and failure strata retained

A Development inputs

SCDD: 351 train + 120 val
UCF: 430 train + 144 val
Source masks inform dense fits
Validation selects checkpoints, rates, temperature and the image operating threshold

B Complete recipes

Frozen DINOv3 / LoRA / full
Adapted CNNs and local ports
Classical, reference and optimization controls
427 families; 1,061 views

C Frozen once

Original weights and rules
1,683 prediction jobs
6,212 endpoint contrasts
Fixed populations and cases
No target-driven changes

D Saved predictions

SCDD: 117 images; 1,061 jobs
UCF: 507 images; 311 jobs
PVEL: 12,326 images; 311 jobs
Same saved PVEL scores yield primary 12,292 + sensitivities

Study-wide design exposure

Both UCF and SCDD development informed the study.
86 UCF training + 144 validation masks are inside 574 HPO images.
18 SCDD training masks at 5% require 120 validation masks too.
Sample-disjoint prediction does not erase research-selection exposure.

Complete analysis and narrower inference

Matching held-source SCDD strata: ARTS 43 / CFVS 27 / CSIR 47.
All declared contrasts and both frozen secondary policies.
Conditional, unadjusted family intervals; pooled pixel AP point-only.
PVEL boxes cannot support mask Dice or a capture-family CI.

Separate evidence: 97 source cost cases; ELPV: 24 outer fits / 480 inner-grid fits on 1,968 development images.

Views and deterministic reuse are not new training replications. Physical modules and pretraining membership remain unverified.

This diagram summarizes the frozen design. It contains no predictive results, model ranking or added scientific choice.